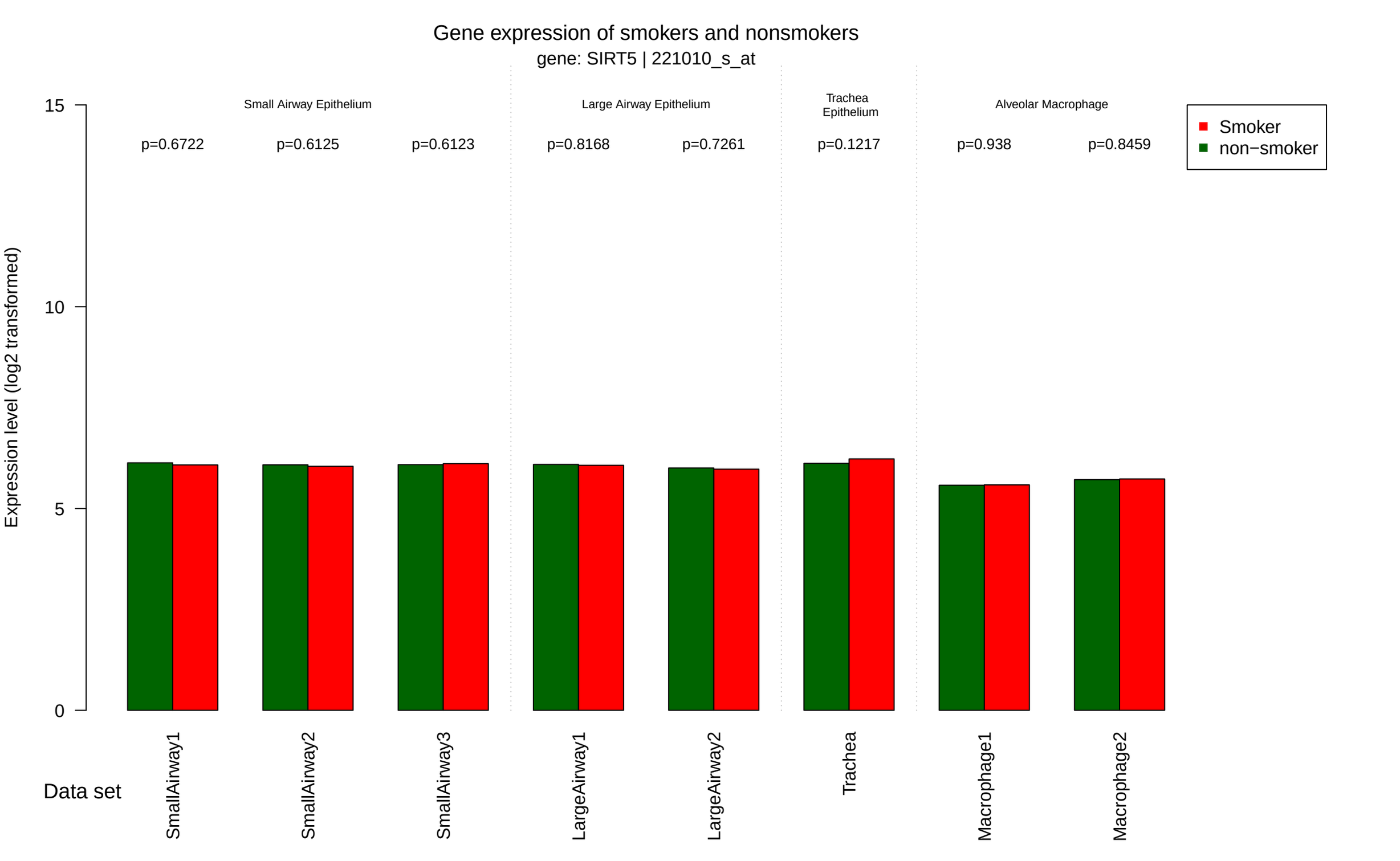
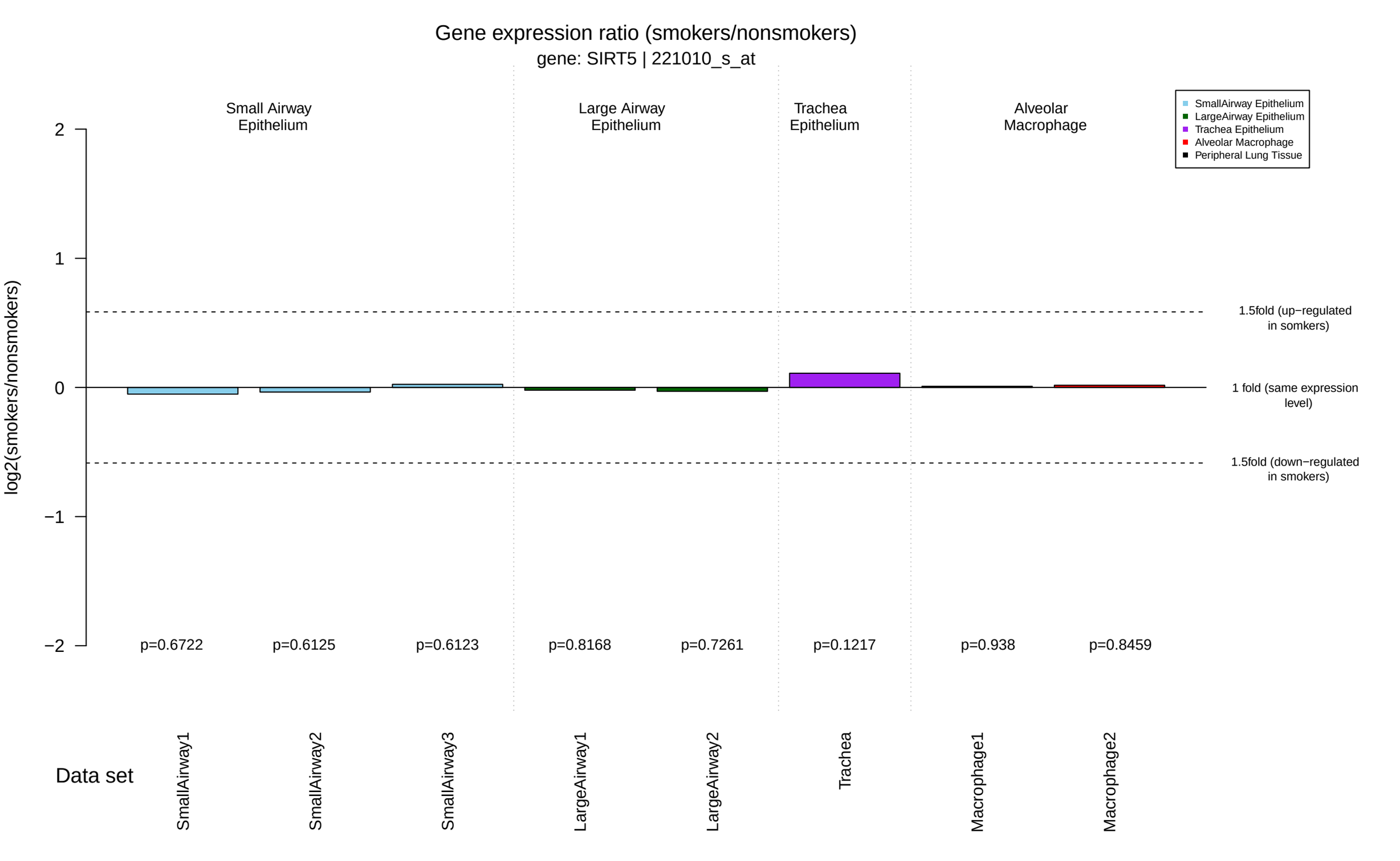
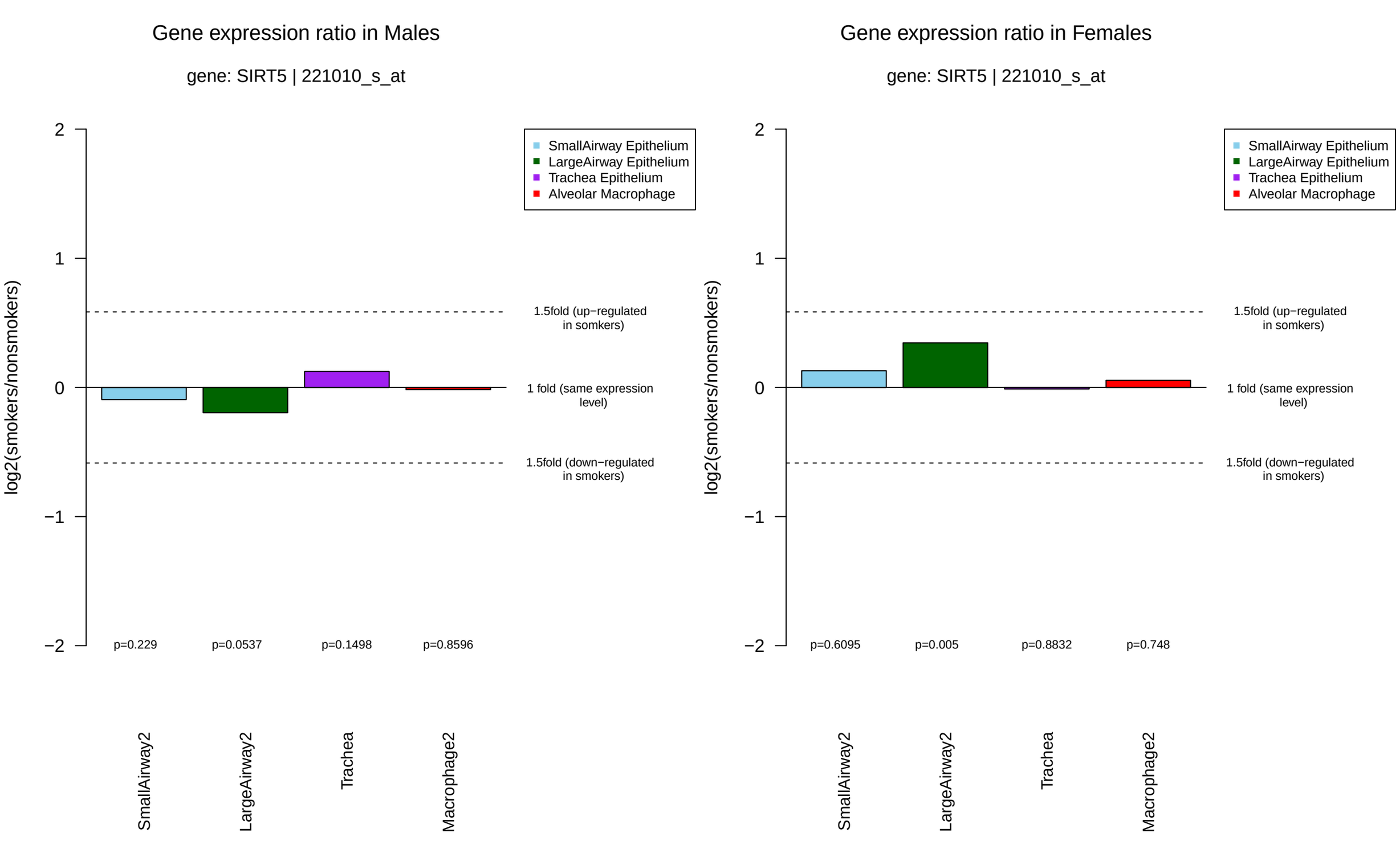
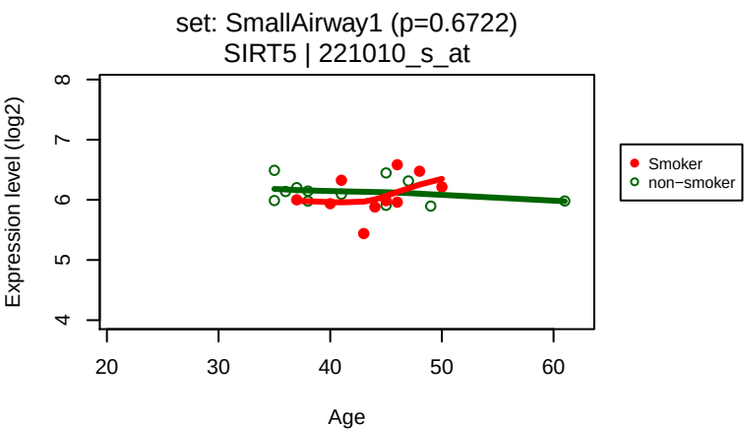
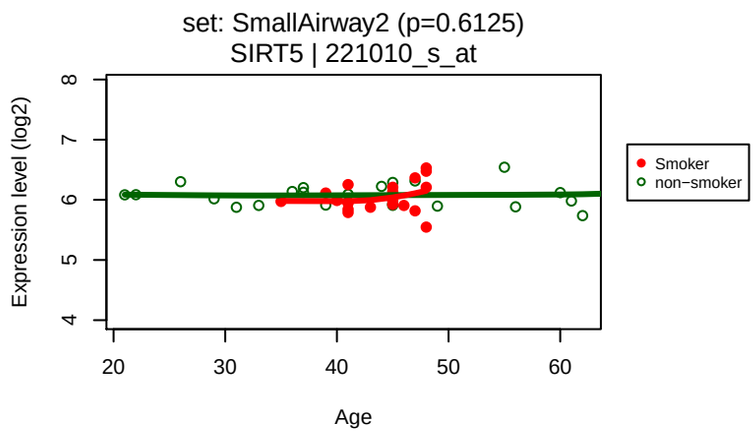
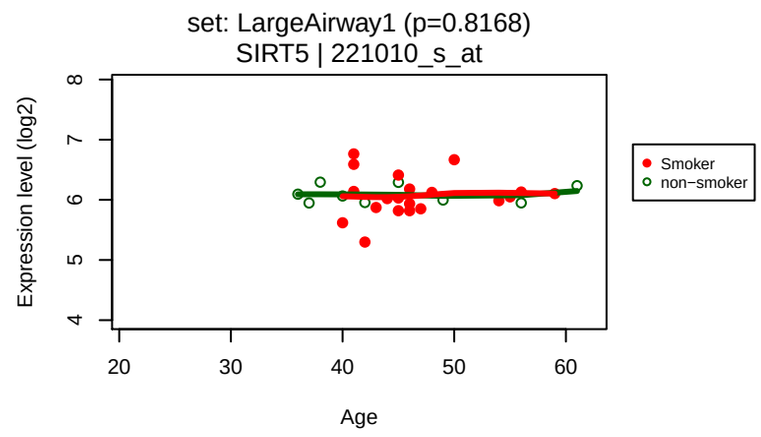

Gene expression of smokers and nonsmokers
gene: SIRT5 | 221010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6722
p=0.6125
p=0.6123
p=0.8168
p=0.7261
p=0.1217
p=0.938
p=0.8459
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

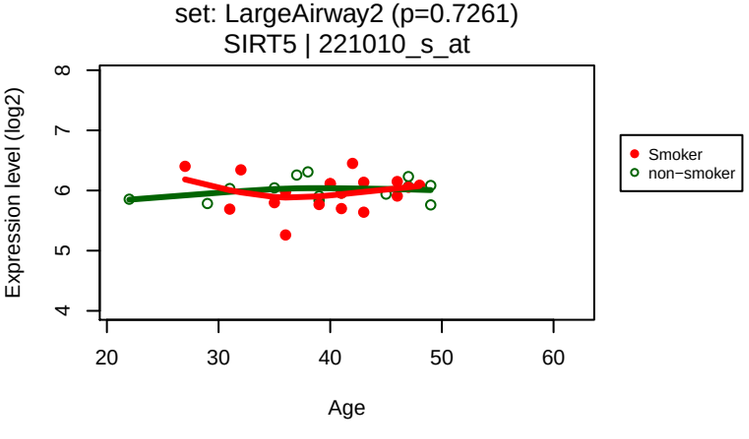

Gene expression ratio (smokers/nonsmokers)
gene: SIRT5 | 221010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6722
p=0.6125
p=0.6123
p=0.8168
p=0.7261
p=0.1217
p=0.938
p=0.8459
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

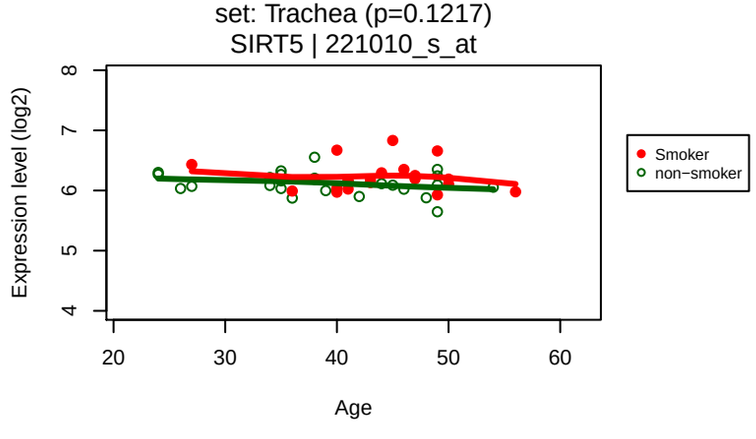

Gene expression ratio in Males
gene: SIRT5 | 221010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.229
p=0.0537
p=0.1498
p=0.8596
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT5 | 221010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6095
p=0.005
p=0.8832
p=0.748
SmallAirway2
LargeAirway2
Trachea
Macrophage2

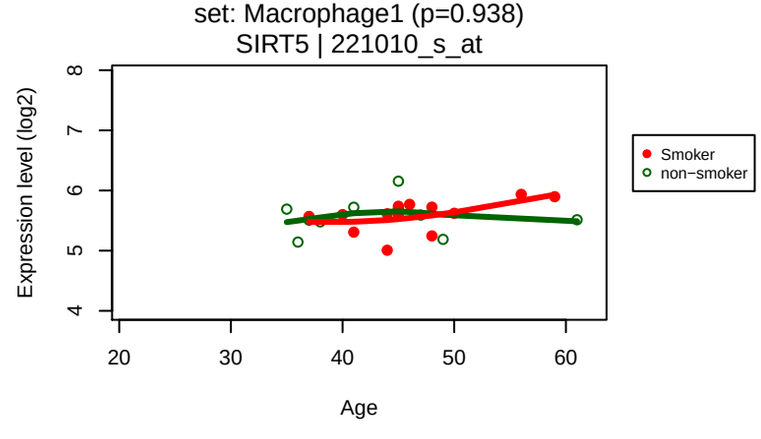

set: SmallAirway1 (p=0.6722)
SIRT5 | 221010_s_at
Expression level (log2)
Age
Smoker
non−smoker

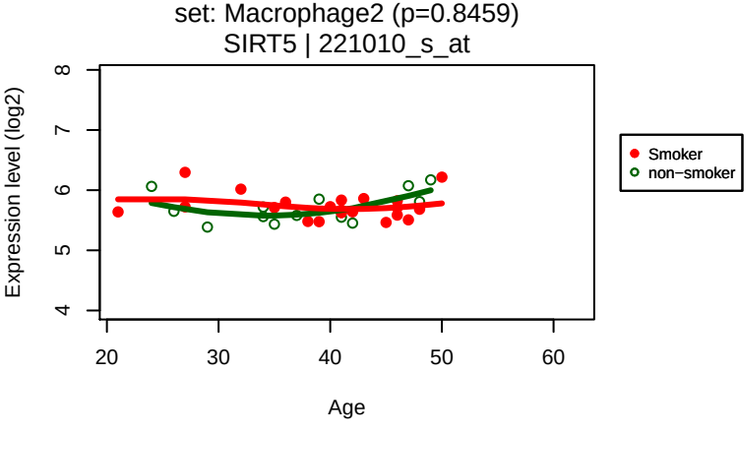

set: SmallAirway2 (p=0.6125)
SIRT5 | 221010_s_at
Expression level (log2)
Age
Smoker
non−smoker

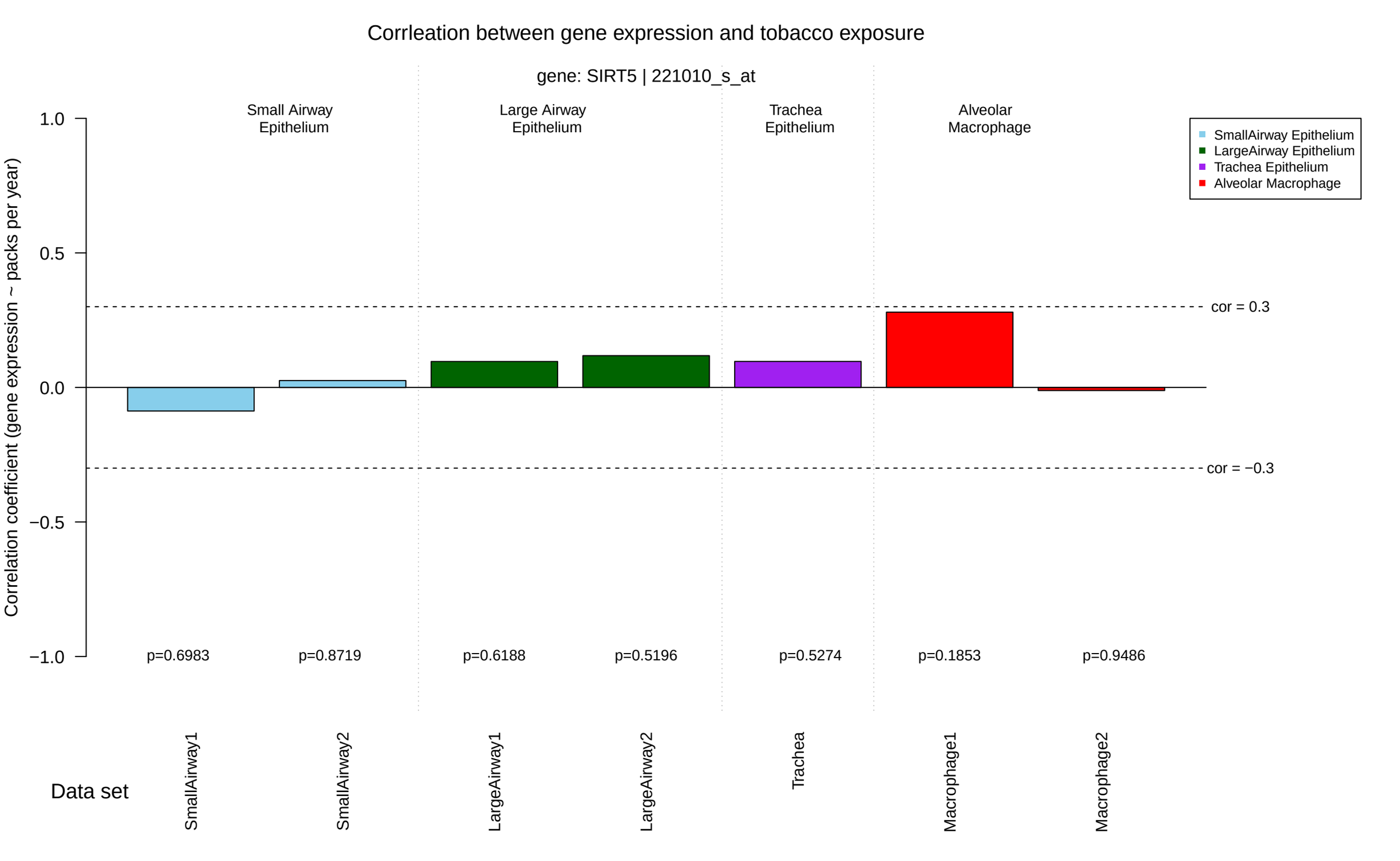

set: LargeAirway1 (p=0.8168)
SIRT5 | 221010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7261)
SIRT5 | 221010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1217)
SIRT5 | 221010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.938)
SIRT5 | 221010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8459)
SIRT5 | 221010_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT5 | 221010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6983
p=0.8719
p=0.6188
p=0.5196
p=0.5274
p=0.1853
p=0.9486
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2